Theory Basics - 1

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The **Closest Pair of Points** problem is a famous problem in the field of **Computational Geometry**.

Given we have n points in a metric space, we need to find the pair of points having the smallest distance between them.

The first solution that comes to mind is the **Brute-force** approach. However, it is not very efficient, having a time complexity of $\mathbf{O(n^2)}$ in **Big-O notation**. This can be improved if we try to.

But let us first understand a few basic concepts related to the problem.

• Compact Sets

Compact Sets are a fundamental concept in **Topology** and analysis. In the context of **metric spaces** or **topological spaces**, a compact set is one that has the property that every **Open Cover** (A collection of open sets whose union contains the set) has a **Finite Sub-cover** (a finite subset of these open sets whose union still contains the set).

Here are some key characteristics and properties of Compact Sets:

1. **Finite Sub-cover Property**: A set K in a topological space X is compact if for every open cover $\{U_\alpha\}_{\alpha \in A}$ of K (where A is an indexing set), there exists a finite subset $\{U_{\alpha_1}, U_{\alpha_2}, \ldots, U_{\alpha_n}\} \subseteq \{U_\alpha\}_{\alpha \in A}$ such that $K \subseteq U_{\alpha_1} \cup U_{\alpha_2} \cup \cdots \cup U_{\alpha_n}$.
2. **Closed and Bounded (in Euclidean Spaces)**: In $\mathbb{R}^n$ (Euclidean space), a set is compact if and only if it is closed and bounded. This is known as the **Heine-Borel theorem**.
3. **Generalization to Arbitrary Spaces**: In general topological spaces, compactness is defined using the open cover criterion rather than boundedness and closure. A space in which every set is compact is called a compact space.

4. **Examples**:

   - In $\mathbb{R}$, a closed interval $[a, b]$ is compact.
   - In $\mathbb{R}^n$, a closed and bounded subset is compact.
   - In more abstract spaces, such as function spaces with the right topology, there are other compact sets.

Silver Triangles

Silver Triangles are the triangles whose area is very very less as compared to their circumcircle.
It can also be stated as:
The Silver triangles have one angle very less that the other two, making the area really very small which is non-comparable to its circumcircle.

• Convex Hull

Convex Hull, also known as **Convex Envelope** or **Convex Closure** of a shape is the smallest convex set containing it. It can also be defined as the intersection of all convex sets containing a given subset of a Euclidean Space.

For a bounded subset of the plane, the convex hull may be visualized as the shape enclosed by a rubber band stretched around the subset.

Convex Hulls of open sets are open, and that of **Compact Sets** are compact. Every compact convex set is the Convex Hull of its extreme points.

Topologically, the convex hull of an open set is always itself open, and the convex hull of a compact set is always itself compact. However, there exist closed sets for which the convex hull is not closed. For instance, the closed set

$$\left\{ \{(x, y) \mid y \geq \tfrac{1}{1+x^2}} \right\}$$

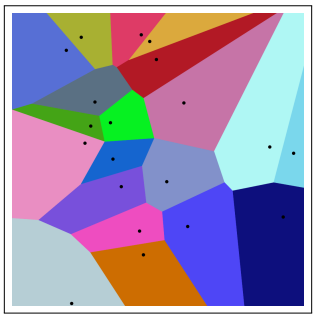
(the set of points that lie on or above the **Witch of Agnesi** has the open upper half plane as its Convex Hull).

• Voronoi Diagram

A **Voronoi Diagram** (also called **Dirichlet Tessellation**) is a type of **Tessellation** pattern in which a number of points scattered on a plane subdivides it into exactly n cells, each

enclosing a portion of the plane that is closest to each point. The points are referred to as **seeds** and its corresponding region is called a **Voronoi Cell** or a **Thiessen Polygon**.

- The **Dual Graph** for a Voronoi diagram (in the case of a Euclidean space with point sites) corresponds to the **Delaunay Triangulation** for the same set of points.
- Assume the setting is the Euclidean plane and a discrete set of points is given. Then two points of the set are adjacent on the convex hull if and only if their Voronoi cells share an infinitely long side.
- The closest pair of points corresponds to two adjacent cells in the Voronoi Diagram.



Voronoi Diagram

Examples

- A **simple cubic lattice** gives the cubic honeycomb.
- A **hexagonal close-packed lattice** gives a tessellation of space with trapezo-rhombic dodecahedra.
- A **face-centred cubic lattice** gives a tessellation of space with rhombic dodecahedra.
- A **body-centred cubic lattice** gives a tessellation of space with truncated octahedra.

• Delaunay Triangulation

In computational geometry, a **Delaunay triangulation** or **Delone triangulation** of a set of points in the plane subdivides their convex hull into triangles whose circumcircles do not contain any of the points. This maximizes the size of the smallest angle in any of the triangles, and tends to avoid **sliver triangles**.

The Delaunay triangulation of a discrete point set P in general position corresponds to the dual graph of the Voronoi diagram for P. The circumcenters of Delaunay triangles are the

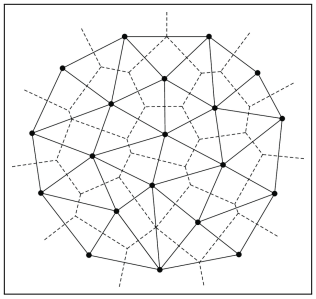
vertices of the Voronoi diagram. In the 2D case, the Voronoi vertices are connected via edges, that can be derived from adjacency relationships of the Delaunay triangles: If two triangles share an edge in the Delaunay triangulation, their circumcenters are to be connected with an edge in the Voronoi tessellation.

Special cases where this relationship does not hold, or is ambiguous, include cases like:

- Three or more collinear points, where the circumcircles are of infinite radii.
- Four or more points on a perfect circle, where the triangulation is ambiguous and all circumcenters are trivially identical.

Properties

1. The union of all **Simplices** in the triangulation is the convex hull of the points.
2. In the plane, the Delaunay triangulation maximizes the minimum angle. Compared to any other triangulation of the points, the smallest angle in the Delaunay triangulation is at least as large as the smallest angle in any other.
3. A circle circumscribing any Delaunay triangle does not contain any other input points in its interior.
4. If a circle passing through two of the input points doesn't contain any other input points in its interior, then the segment connecting the two points is an edge of a Delaunay triangulation of the given points.
5. The closest neighbor b to any point p is on an edge PB in the Delaunay triangulation since the **Nearest Neighbor Graph** is a sub-graph of the Delaunay triangulation.



Delaunay Triangulation with Voronoi Diagram

In two dimensions, the following determinant detects whether a point D lies within the circumcircle of the points A, B and C:

$$\begin{vmatrix} A_x & A_y & (A_x)^2 + (A_y)^2 & 1 \\ B_x & B_y & (B_x)^2 + (B_y)^2 & 1 \\ C_x & C_y & (C_x)^2 + (C_y)^2 & 1 \\ D_x & D_y & (D_x)^2 + (D_y)^2 & 1 \end{vmatrix}$$

When A, B and C are taken in counter-clockwise order, this determinant is positive only if D lies inside the circumcircle.

• Divide and Conquer Algorithm

The **Divide and Conquer algorithm** for the closest pair problem works as follows:

1. **Divide:** Split the set of points into two halves by a vertical line.
2. **Conquer:** Recursively find the closest pair of points in each half.
3. **Combine:** Determine the closest pair of points that straddles the dividing line and compare it with the closest pairs found in the two halves.

Algorithm Steps:-

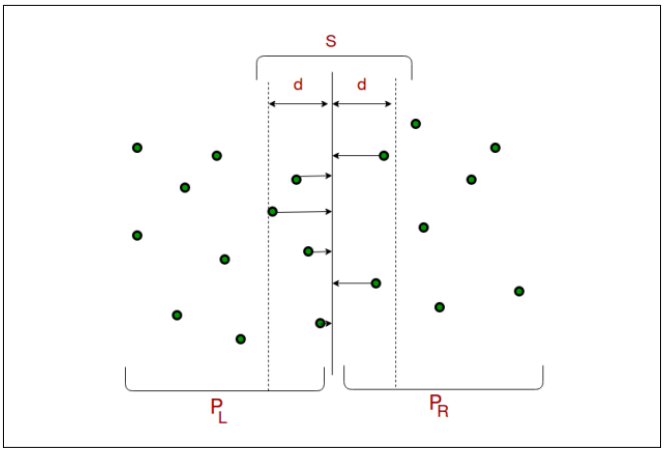
Let P be the set of points and let P_x and P_y be the sets of points sorted by their x-coordinates and y-coordinates, respectively.

1. **Base Case:** If the number of points is small (e.g., 2, 3 or 4), solve the problem using the brute force method.
2. **Divide:** Split P_x into two halves P_L and P_R around the median x-coordinate.
3. **Recursion:** Recursively find the closest pairs in P_L and P_R, denoted as d_L and d_R. This step involves applying the same Divide and Conquer approach to each of the two halves of the set of points. In essence, we're solving the closest pair problem independently for each half.

The job is still half done.

We will now combine the results.

- Find the minimum distance d = min(d_L, d_R).



Closest Pair Algorithm

- Now we want to see if there exists a closer pair made by picking one point from the left part and one from the right part.
- Consider a band of width 2d as shown in the figure.
- Identify the points that lie within this band. Sort these points by their y-coordinates to form S.
- Iteratively Check the distances between each point in S and the six points below that point. (**WHY SIX!?**)
- Update d if a smaller distance is found.

Conclusion

The problem of finding the **closest pair of points** is a fundamental challenge in computational geometry. While the **brute-force** method offers a straightforward solution, it is computationally expensive for large datasets. By understanding and applying advanced geometric concepts such as **convex hulls**, **Voronoi diagrams** and **Delaunay triangulation**, we can significantly improve the efficiency of our algorithms. The divide and conquer approach, in particular, provides a powerful method to solve the closest pair problem in **logarithmic time**, making it a crucial technique for handling large-scale geometric data efficiently.
